

EXERCISE 2

SQL AGGREGATES & OPERATORS

→ TABLE STUDENTS

Student ID	Name	Age	Department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q1. List all distinct departments in the student table

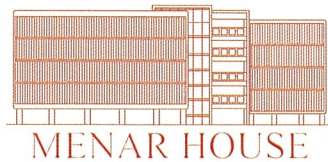
1 SELECT DISTINCT department

2 FROM student;

3

Results

Department
IT
HR
Finance



Q2 Average age per department

1. `SELECT` department `AVG(age)` `AS` avg_age
2. `FROM` students
3. `GROUP BY` departments;

Results

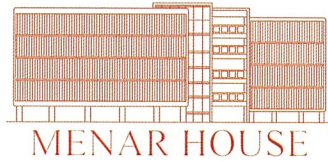
department	avg_age
IT	20.5
HR	22
Finance	23

Q3 Departments with more than 1 student

1. `SELECT` department, `COUNT(*)` `AS` Student_Count
2. `FROM` students
3. `GROUP BY` department
4. `HAVING` `COUNT(*)` `>` 1;
5. `""`

Results

department	Student_Count
IT	2
HR	2



Q4. Students aged between 21 and 23

```
1 SELECT student_id, name, age, department
2 FROM students
3 WHERE age BETWEEN 21 AND 23;
```

4
Results

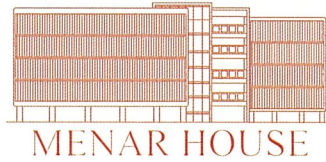
Student_id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5 IT or HR students older than 21

```
1 SELECT student_id, name, age, department
2 FROM students
3 WHERE (department IN ('IT', 'HR')) AND
age > 21;
```

Results

Student_id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR



02 The course Table

→ TABLE courses

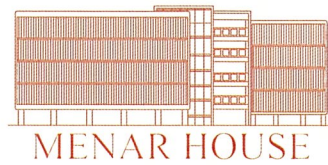
course id	Course name	department	credits
101	SQL Basic	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

b Total credits > 5 per department

- 1 SELECT department, SUM(credits) AS total_credits
- 2 FROM Courses
- 3 GROUP BY department
- 4 HAVING SUM(credits) > 5;

Results

department	total_credits
IT	11



07 Courses not equal to 4 credits

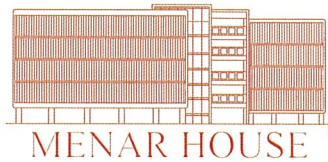
```
1 SELECT course_id, course_name, depart, credits
2 FROM courses
3 WHERE credits <> 4;
4 "
```

Results

Course_id	Course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08 Top 3 courses by credits

```
1 SELECT course_id, course_name, credits
2 FROM courses
3 ORDER BY credits DESC
4 LIMIT 3;
```

03 The enrollments Table

→ TABLE enrollments

enrollment-id	student-id	course-id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

09 Max, Min, Avg grade

1 SELECT

2 MAX (grade) AS max_grade,

3 MIN (grade) AS min_grade,

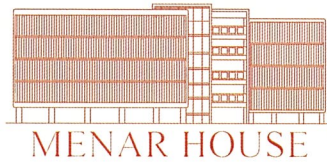
4 AVG (grade) AS avg_grade

5 FROM enrollment;

6

Results

Max-grade	Min-grade	avg-grade
90	78	84.6

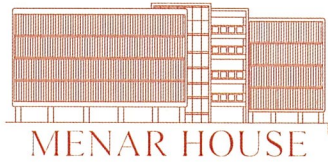


10. Enrollments per course

```
1 SELECT course_id, COUNT(*) AS enrollment_count
2 FROM enrollments
3 GROUP BY course_id;
4 ''
```

Results

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1



04 The Salaries table

→ TABLE Salaries

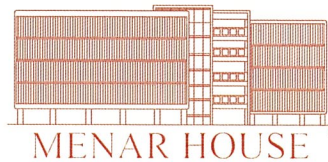
employeeid	Name	department	Salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11. Total salary & bonus per department

```
1 SELECT
2     department
3     SUM(Salary) AS total_salary,
4     SUM(bonus) AS total_bonus
5 FROM Salaries
6 GROUP BY department;
```

Results

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000



12. Departments with avg salary > 55000

```

1 SELECT department, AVG (Salary) AS avg-Salary
2 FROM Salaries
3 GROUP BY department
4 HAVING AVG (Salary) > 55000;

```

Results

department	avg-Salary
IT	61000
Finance	70000

13 Employees with total compensation > 60000

```

1 SELECT
2     employee_id,
3     name,
4     salary,
5     bonus,
6     (Salary + bonus) AS total_compensation
7 FROM Salaries
8 WHERE (Salary + bonus) > 60000

```

employee_id	Name	Salary	bonus
1	Tom	60000	5000
3	Spike	70000	6000
4	Tyke	62000	5500



05 PROJECT TABLE

→ TABLE projects

project_id	project_name	department	budget
1	AI App	IT	12 000
2	Paysroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

14. Total & avg budget (avg > 70 000)

1 SELECT

2 department

3 SUM(budget) AS total_budget

4 AVG(budget) AS avg_budget

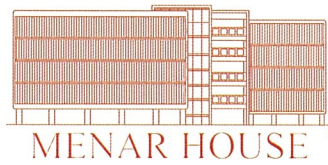
5 FROM projects

6 GROUP BY department

7 HAVING AVG(budget) > 70 000;

Results

department	total_budget	avg_budget
IT	27 000	135 000
Finance	80 000	80 000



15. Projects between 50k-120k
excluding Marketing

```
1 SELECT project_id, project_name, dept, budget
2 FROM Projects
3 WHERE budgets BETWEEN 50000 AND 120000
4 AND department <> 'marketing';
```

Results

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	HR Portal	HR	50 000
5			